

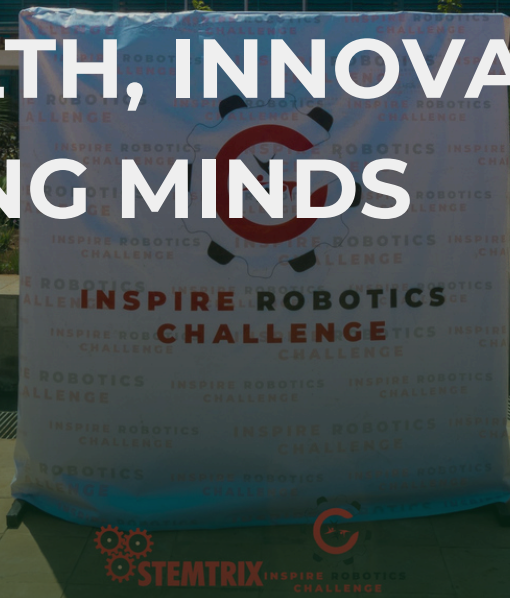
#IRC2025RECAP

IRC 2025

SECOND EDITION INSPIRE ROBOTICS
CHALLENGE



IRC 2025: A CELEBRATION OF HEALTH, INNOVATION, AND YOUNG MINDS



The much-anticipated **inspire robotics Challenge (IRC) 2025** was a resounding success, bringing together bright young minds for a day of learning, creativity, and innovation. Held on **15th March 2025** at the prestigious **Porterhouse Senior School, Runda**, the event saw enthusiastic participation from 20 schools, each showcasing their unique perspectives on this year's theme: **Health**

With a focus on fostering curiosity and critical thinking, the competition was thoughtfully divided into three age-appropriate categories:

Explore -Juniors (4–9 years) explored the fascinating world of **Germs**, delving into how these tiny organisms affect our daily lives and the importance of hygiene in staying healthy.

Challenge-(10–13 years) took on the challenge of understanding **Diseases**, investigating causes, prevention, and the science behind treatments.

Innovation-(14–17 years) ventured into the cutting-edge realm of Artificial Intelligence in **Healthcare**, examining how technology is revolutionizing medical diagnostics, treatment plans, and patient care.

We extend our heartfelt gratitude to all participating schools, teachers, judges, and supporters who made IRC 2025 an unforgettable experience. Stay tuned as we dive deeper into the highlights, winning projects, and inspiring moments from this remarkable day





4. Brighton School Bright ideas that shine through innovation.

5. Jawabu School-Small thinkers with big, bold solutions.

6. Holy Cross Catholic International School-Faith in science, passion for progress.



APPRECIATION

A STANDING OVATION TO OUR BRILLIANT PARTICIPANTS!

The magic of IRC 2025 wouldn't have been possible without the vibrant minds and tireless enthusiasm of these incredible schools. Each one brought unique ideas, energy, and innovation to the competition—turning theory into action and challenges into solutions.

Let's roll out the applause for

1. Riara Group of Schools-Where curiosity meets excellence.
2. Potterhouse Runda-Our gracious hosts and fierce competitors.
3. Brookhouse Rundal-Blending creativity with cutting-edge thinking.
7. Sabis International-Precision and passion in every project.
8. St Andrews Turi-Global perspectives, local impact.
9. Braeburn International School-Where STEM sparks fly.
10. Cybertech-Tech wizards shaping tomorrow's healthcare.
11. Stemplens-Sharp focus on scientific breakthroughs.
12. Premier Academy- Where hypotheses become realities.
13. International School of Kenya-Innovation without borders.
14. Children's Garden School-Tiny hands building big futures.
15. Braeburn School Thika- Proof that great ideas bloom everywhere.
16. Brookhouse Karen-Creativity in motion.
17. French School-Smartness in innovation.

To every student, teacher, and mentor who made **IRC 2025** unforgettable-thank you. Your brilliance didn't just compete; it inspired. We can't wait to see how you'll heal the world.

PROJECTS

IRC 2025: INNOVATING HEALTHCARE THROUGH YOUNG MINDS

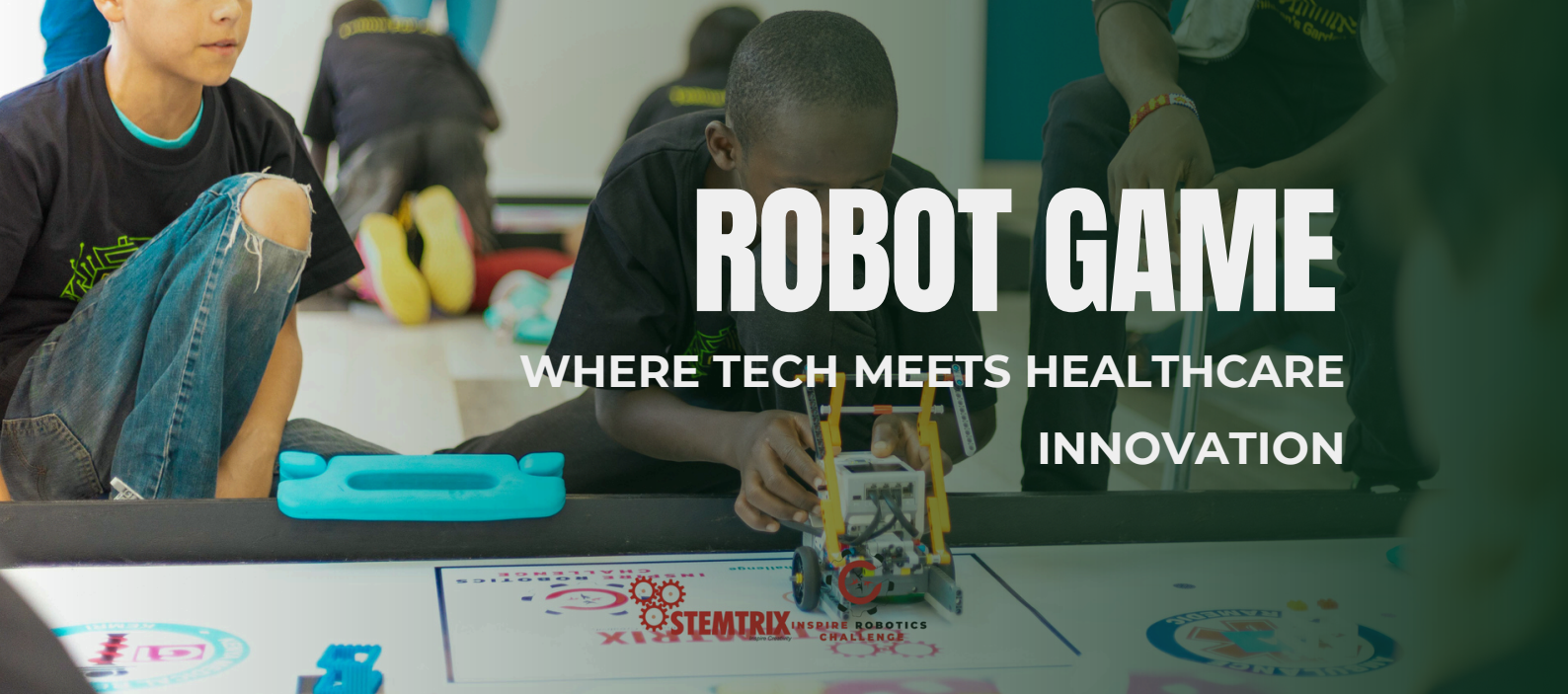
IRC 2025: PROJECTS OVER VIEW

Kenya's healthcare challenges inspired this year's IRC theme-Health- chosen by **STEMTRIX** to spark tech-driven solutions from young innovators. Over three months, students from 20 schools immersed themselves in real-world problems affecting the Kenyan economy, researching key areas based on their age groups. Juniors explored germs and hygiene, seniors tackled disease prevention and treatment, while older students delved into AI applications in healthcare. Their mission was clear: develop creative, practical solutions to improve healthcare access and efficiency.

The students took full ownership of their projects, with teachers acting only as facilitators-not researchers or decision-makers. This rule ensured that every prototype, from low-cost diagnostic tools to AI-powered health bots, was entirely student-driven. There were no limits on creativity, allowing teams to explore bold, unconventional ideas. Some focused on community education, while others engineered devices to address gaps in medical resources. The diversity of solutions proved that when young minds are given freedom, innovation flourishes.



A highlight of IRC 2025 was the robot challenge, where teams programmed robots to perform healthcare tasks like vaccine delivery, patient monitoring, and AI-assisted diagnostics. This hands-on competition reinforced the link between technology and medicine, sharpening skills in coding, engineering, and problem-solving. Students learned to troubleshoot in real time, adapting their robots to simulated medical emergencies-an experience that mirrored real-world healthcare challenges. Beyond robotics, the event showcased how interdisciplinary collaboration can drive change. Teams blended biology, engineering, and data science to create holistic solutions. One group designed a germ-tracking app for schools, while another built a portable diagnostic kit for rural clinics. The projects didn't just exist in theory—they addressed actual pain points in Kenya's healthcare system, proving that student-led innovation has real-world impact.



ROBOT GAME

WHERE TECH MEETS HEALTHCARE
INNOVATION



The Robot Game brought this year's health theme to life in the most thrilling way! More than just a competition, it was a *real-world simulation* of how robotics and AI can revolutionize healthcare. Students became *engineers, coders, and problem-solvers*, designing and programming robots to tackle medical challenges—just like future innovators would in hospitals, labs, or emergency response.

Mission Possible: Coding for a Healthier Future
Before the big day, **STEMTRIX** tutors guided learners through hands-on robotics training, equipping them with skills to:

Design and build efficient robot prototypes
Program precise movements to complete healthcare missions
Troubleshoot in real-time like true engineers

On competition day, teams faced a **2.5**-minute timed challenge with multiple missions:

1. Emergency medical delivery simulations
2. Precision vaccine transport tasks
3. AI-assisted diagnosis scenarios

Each successfully completed mission earned points, with judges carefully evaluating:

- ✓ Technical execution
- ✓ Creative problem-solving
- ✓ Adherence to healthcare applications

The Electric Atmosphere

The arena buzzed with excitement as **parents and teachers cheered** every strategic move. The two-round format doubled the intensity—teams refined strategies between rounds, showing incredible adaptability.

This wasn't just about winning; it was about proving how robotics can save **lives**. From the nervous first clicks of code to the triumphant final missions, every moment showed Kenya's next generation of health-tech pioneers in action.





JUDGES TAKE

VERDICT: RISING STARS OF STEM AND HEALTHCARE INNOVATION



The judging panel—comprised of leading experts from Kenya's healthcare and tech sectors—had one unanimous takeaway: This year's competition raised the bar to extraordinary heights.

"A New Standard of Excellence"

After evaluating the projects and robot challenges, the judges noted a remarkable leap in quality compared to previous years.

Dr. Joel Lessan, Paediatric Surgeon and a parent attendee, commended Stemtrix for organizing such an impactful event. He emphasized the importance of initiatives like these in nurturing problem-solving minds from an early age and guiding young learners toward future career paths. 'Efforts like these are vital in shaping the next generation of innovators and thinkers,' he noted.

Most of the judges were impressed with the dedication from schools. We need even more institutions to join these challenges transform textbook knowledge into life-changing skills. They noted

The Toughest Decisions Yet

Judge Hawi Rapudo-Team Lead Kijiji Yetu confessed this was their most challenging year to score:

- Every robot game run had multiple flawless executions
- Research projects blended creativity with viable healthcare solution
- The two-round format revealed teams' ability to adapt and improve

Elias Mutua, Head of Computing (Prep), APP Coordinator, Year 8 Kili Boys Tutor, St Andrew's Preparatory School, Turi remarked: We wished we could award every team-the gap between 'good' and 'excellent' became razor-thin.



CELEBRATING EXCELLENCE

MEET THE IRC 2025 CHAMPIONS!



This year's competition reached new heights, but these schools rose above the rest with groundbreaking innovation, technical mastery, and exceptional teamwork. Join us in applauding the brightest stars of IRC 2025—and Kenya's future STEM leaders!

Explore Category

Premier Academy | Brookhouse School | St. Andrews Turi | Sabis International School

The young learners in this category demonstrated exceptional mastery of concepts, particularly in their focus on health-related challenges. From their meticulous work on robot boards to their ability to simplify complex ideas, their presentations revealed deep understanding. Judges were especially impressed by their clarity, creativity, and confidence in tackling abstract topics.

Challenge Category

International School of Kenya |
Potterhouse | Braeburn | Jawabu
School | St. Andrews Turi

These teams stood out for their dedication to research and execution. Every project reflected thorough preparation—from problem-solving to articulate explanations of their challenges. Judges noted the learners' passion and adaptability, as they embraced feedback and refined their work even during presentations. A true testament to growth under pressure.



Innovators Category

Brookhouse School | Jawabu School |
International School of Kenya |
Cybertech

Innovation met expertise in this category, where participants showcased unmatched technical maturity. Their use of industry-standard terminology and project principles highlighted their rigorous preparation. Beyond skill, judges praised their teamwork and core values—collaboration and professionalism shone through every presentation.

Congratulations to All!

Each team embodied the spirit of curiosity and excellence. Stay tuned for next event's even bigger challenges.



A WORD FROM OUR DIRECTOR

CELEBRATING KENYA'S STEM REVOLUTION



As we close this chapter of **IRC 2025**, I am filled with immense pride and gratitude for what we've achieved together.

To the dedicated teachers—your unwavering commitment to guiding these young minds through research, prototyping, and presentation has been nothing short of extraordinary. You didn't just teach; you ignited curiosity and shaped the innovators of tomorrow.

To the remarkable parents—your presence, encouragement, and belief in the power of **STEM** have set a powerful example. By supporting your children's passion for robotics and technology, you're nurturing the architects of Kenya's future.

What we witnessed at IRC 2025 was more than a competition—it was proof of progress. The creativity, technical skill, and problem-solving on display tell me one thing: **Kenya** is undeniably on track to surpass its Vision 2030 goals. These students aren't just learning; they're building the solutions that will transform our healthcare system and beyond.

Today's prototypes are tomorrow's breakthroughs.

Stay connected for 2026 updates! Follow us on Social Media Handles or Visit Website.

This is only the beginning. **STEMTRIX** is doubling down with even more exciting programs, partnerships, and opportunities to bring cutting-edge **STEM** education to every school. Together, we'll ensure Kenya's youth lead the **Global Tech Revolution**.

Thank you for believing in this mission. The future isn't just bright—it's brilliant, bold, and ours to create.

Warm regards,
Christine Onyango
Director, **STEMTRIX** East Africa

